Effects of Container Type and Silica Gel Amount on Cat Food Storage

30 g of cat food per container; 3 independent repeated groups per treatment

Abstract

The experiment is based on the environment in our automatic cat food machine project. It uses loss-on-drying (LoD), pH, and TLC color signals to test lipid-soluble extracts, studying cat food storage under different silica gel amounts in plastic and foil containers. Silica gel lowered the LoD in both containers on D14, but the effect on D21 only remained in the foil groups(0 g, 7.6%; 2 g, 6.67%; 4 g, 4.5%). The pH results show that there was a slightly lower pH in the foil group than in the plastic group across different amounts of Silica gel on D14 (0.05- 0.09), and lower on D21. In the TLC test on D21, the 1 g and 2g groups had significantly higher TLC integrated intensity than the foil control (0 g,3964;1 g,5003;2 g,5787), while plastic showed no difference. The results suggest that moisture control may be related to the storage condition of lipid-soluble components, and adding more silica gel does not provide a significantly higher TLC signal. pH suggests that pH differences may be brought about by the material rather than other factors such as silica gel.

Introduction

There is a major issue when cat food is exposed to air for a long time: the texture of the cat food becomes soft and may turn sour, which can cause dislike in cats. During long-term storage, the cat food will absorb moisture from the near environment, and its antioxidant ability will decrease over time. Lipid nutrients are one of the major nutrients and additives in cat food; these lipids include antioxidants such as vitamin E. Under these conditions, oxidation and an increase in moisture lead to an overall decrease in the quality of cat food.

High moisture makes the cat food easier for mold and bacteria to grow; triglycerides in lipids will be broken down by hydrolysis into fatty acids, which drive the pH down on their own. It also affects the antioxidant ability since lipids react with Oxygen through oxidation reactions; this is also accelerated by factors such as light, heat, and degree of exposure to air. One molecule that contributes a significant role in preventing oxidation, and is an essential nutrient for cats, is Vitamin E, which provides hydrogen to free radicals that protect the chain reaction from lipids. (Ralla et

al.,2024)The protection process also consumes vitamin E over time. However, in this experiment, vitamin E is not able to be determined directly because iodine stains in TLC contain many other lipid-soluble compounds, but it becomes evidence to suggest or predict the concentration of vitamin E and reflect the percentage of cat food that has been preserved.

Automatic feeders are commonly used by people to save their time and energy, especially for people who are not available at home or are on trips. In these cases, people only need to pour cat food into the feeder and control it to feed the cat automatically. However, as the goal of automatic distribution of cat food, it requires cat food in the storage box; the airflow between refills and distribution and the gaps between the lid or the walls cause a new issue to appear with the ability to store for longer use once a large amount of cat food has been exposed to air.

Plastic storage boxes are the most commonly used material in all cat feeder machines. It stands for the lower prices and convenience. Aluminum foil was also tested because it is commonly seen in kitchens and blocks light exposure. Furthermore, silica gel is a desiccant; it can absorb water and keep the near environment dry for a certain period of time.

Investigate the effects of plastic and foil containers with different amounts of silica gel on the storage conditions of cat food, and how this applies to the automatic feeder we designed. There are three questions: whether adding silica gel has an effect; whether more silica gel has better preservation of the cat food; and whether different containers affect the preservation. I hypothesize that more silica gel would better lower moisture and help preserve lipid-soluble components in a plastic container.

Methods

1. Groups and Storage

Use Royal cat food for the entire experiment in two types of containers: plastic and aluminum foil. Each container is treated with 0, 1, 2, or 4 g of silica gel, with 0 as the control. Each treatment has 3 replicate groups with 30g of cat food each, for a total of 8 treatments and 24 containers. The cat food was stored in a basement (~65% humidity, ~22C)

The relative humidity was 74% on Day 0, 64% on Day 14, and 65% on Day20.

2. Loss on Drying(LoD)

The empty aluminum dish was weighed first. About 1g of cat food was added, and the loaded weight of the cat food and dish was weighed before heating M1. The sample was heated to 250°F for 3 h, cooled for 20 m, then weighed as M2. Each container was tested once on each test day.LoD = (M1-M2)/(M1-M0)x 100%.

The higher LoD indicates the larger fraction of mass was lost during the heating. This is used as a water indicator. However, the heating may also remove other volatile substances, it is not able to exactly and accurate to prove the moisture concentration.

3. TLC Lipid- Soluble Signal

0.5 g of cat food was ground and mixed with 2.5 mL of hexane, and the tube was capped for 20min away from light. The extract was separated, and another 2.5 mL of hexane was added, then removed in the same way. The plate was dipped in hexane: ethyl acetate 9:1(). After the treated cat food solvent was dipped on the plate, it was stained with iodine solution and photographed with a phone; ImageJ was used to calculate the integrated intensity of the selected spot area, which reflects the color intensity, but this can't be used as a tool to determine the concentration of Vitamin E or total fat remaining.

Each group had 6 lanes on the plate. Adjacent lanes came from the same container for replicate groups. Therefore, during the data analysis, the data were averaged and tested with 3 independent, same-condition groups. Lane 6 of the D14 plastic 1g group was aberrant; it was excluded from the data.

4. PH

1g cat food for each container was mixed with 10ml of distilled water, stay for 20min. The Ph of water extract was then measured with PH meter with 7PH buffer. Each sample was read twice, and the two readings were averaged; the electrode was rinsed and wiped after each test. pH was measured on D0, D14, and D21; the plastic group was measured first, then the foil groups. Since the calibration record is not stable, the data can only be compared within the same day.

5. Statistics

Each treatment had 3 independent storage container, teh analysis use n=3 , in TLC the adjacent two lanes were averaged. LoD, each container was measured once. Silica gel groups were compared with 0g group of the same type of container on the same day, and the foil was compared with plastic at the same dose. By using Welch's

t-test. P<0.05 was considered significant; not significant means there was not enough information to support differences.

TLC plates are used on different days; it is suggested in this experiment to compare only with the results on the same day.

Results

LoD

Table 1. Mean LoD of each group and comparison with control

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Treatment	D0 LoD (%)	D14 LoD (%)	D14 p	D21 LoD (%)	D21 p
Plastic 0 g	4.50	7.18	—	7.13	—
Plastic 1 g	7.43	3.05	<0.001	7.27	0.730
Plastic 2 g	6.23	6.20	0.016	7.27	0.735
Plastic 4 g	6.00	5.35	0.004	7.10	0.939
Foil 0 g	7.70	8.06	—	7.60	—
Foil 1 g	6.20	5.73	0.003	6.70	0.027
Foil 2 g	3.87	1.76	<0.001	6.67	0.023
Foil 4 g	3.77	2.30	<0.001	4.50	<0.001

n = 3 containers. p: Welch t-test

- LoD was calculated from the weighing records and rounded to two decimals; P is compared with 0g groups.
- D14, all silica gel groups had significantly lower LoD than their own controls, while the foil silica gel groups were significantly lower.

TLC lipid- soluble signal

Table2

Table 2. TLC signal by container and test results				
Treatment	D14 mean signal ± S	D14 p	D20 mean signal ± S	D20 p
Foil 0 g	11,496 ± 2,320	—	3,964 ± 96	—
Foil 1 g	14,190 ± 758	0.173	5,003 ± 414	0.043
Foil 2 g	14,316 ± 929	0.159	5,787 ± 344	0.008
Foil 4 g	12,784 ± 2,136	0.518	5,345 ± 1,001	0.138
Plastic 0 g	12,049 ± 914	—	5,012 ± 670	—
Plastic 1 g	13,623 ± 338	0.082	4,380 ± 550	0.278
Plastic 2 g	13,829 ± 373	0.062	4,787 ± 632	0.695
Plastic 4 g	16,257 ± 511	0.005	4,564 ± 634	0.448
n = 3 containers (adjacent lanes averaged). p: Welch t-test				

- D14, all foil silica gel groups had higher means, and the control(0g) group varied between containers. Plastic groups: only 4g is significantly higher than the control; D20, foil is respectively higher than the control.
- The effects of 2 g didn't remain high in 4g.

TLC Container Comparison

Table3

Table 3. D20 foil groups compared with plastic groups at the same dose			
Silica gel	Foil vs. plastic mean	p	
0 g	-21%	0.111	
1 g	+14%	0.197	
2 g	+21%	0.093	
4 g	+17%	0.328	
Positive - foil mean was higher. p: Welch t-test, n = 3 containers.			

- The foil preserves the lipid-soluble compounds better than Plastic
- D0 should not be considered because it is the initial test; it is the error.

PH

Table 4.

Table 4. Mean pH of each group and foil vs. plastic comparison at the same dose					
Treatment	D0 pH	D14 pH	D14 p	D21 pH	D21 p
Plastic 0 g	5.85	5.80	—	5.84	—
Plastic 1 g	5.86	5.79	—	5.83	—
Plastic 2 g	5.87	5.76	—	5.79	—
Plastic 4 g	5.86	5.77	—	5.79	—
Foil 0 g	5.87	5.71	<0.001	5.79	0.009
Foil 1 g	5.87	5.71	0.006	5.76	0.197
Foil 2 g	5.87	5.71	0.013	5.80	0.877
Foil 4 g	5.85	5.72	0.093	5.76	0.024
n = 3 containers (two readings averaged per container). p: Welch t-test, foil : plastic					

- D0, there was no pH difference between the two containers (P=0.29). On D 14, the foil groups had about 0.065 lower pH than the plastic (P<0.001); it is significant at 0,1, 2g close to significant for 4g.
- D21. The foil is lower (P < 0.006); days 0 and 4 have no significant differences. The consistent effect on pH.
- The drop and rise in pH are the reason for errors in calibration records; the comparison and results across days may not be reliable, but the data from the same day can be used to refer to the effect of different containers and silica gel.

Conclusions

Silica gel can control the moisture in the first two weeks , it's effect get reduced in significantly on D14. It depends on the ratio between the silica gel and the cat food, as well as the house humidity, because it can become saturated quickly if the humidity is high. Foil can better preserve the lipid-soluble signal; it was 26%, 35%, and 46% higher than the control. This may also be due to the lower LoD later on; longer-lasting moisture control may help preserve some lipid-soluble components. The plastic container didn't have the advantage of preserving the TLC signal. The increase in the early stage didn't show up in the last stage, which corresponds to the lower LoD; also find the relationship between the moisture and lipid-soluble compounds preservation. The foil groups have a lower pH than the plastic groups, regardless of silica gel amount. There was no difference on D0, but D 14 and D 20 the foil groups overa were significantly lower than the plastic groups on the same day（0.05-0.08）, The foil groups with silica gel kept the

moisture the lowest, but their PH as low as the foil without silica, so the pH may be comes form container itself, not moisture.